

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomised, Double-blind Study to Evaluate the Effect of Balcinrenone/Dapagliflozin, Compared With Dapagliflozin, on the Risk of Heart Failure Events and Cardiovascular Death in Patients With Heart Failure and Impaired Kidney Function **Short Title/Acronym:** BalanceD-HF **Protocol Number:** D6402C00012 **NCT Number:** NCT06307652 **Sponsor Name:** AstraZeneca **Investigational Product:** Balcinrenone/dapagliflozin (fixed-dose combination capsule) and Dapagliflozin (tablet) **Phase of Development:** Phase III **Medical Monitor:** Arbab Bhatti (AstraZeneca AB) / Dr. Mark Belkin (Lead Physician)

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To determine whether the balcinrenone/dapagliflozin combination is superior to dapagliflozin alone in reducing the risk of cardiovascular (CV) death and heart failure (HF) events with and without hospitalization. * **Secondary Objectives:** To evaluate the safety, tolerability, and efficacy of balcinrenone/dapagliflozin compared with dapagliflozin in patients with chronic HF across the full range of left ventricular ejection fraction (LVEF) and impaired kidney function.

2.2 Background & Rationale Heart failure affects millions worldwide and continues to be a major cause of death and hospitalization. Mineralocorticoid receptor antagonists (MRAs) and SGLT2 inhibitors independently reduce the progression of heart failure and chronic kidney disease via different mechanisms, expecting to exhibit complementary benefits. However, MRAs are significantly underused in patients with reduced kidney function due to safety concerns like hyperkalemia and an initial reduction in eGFR. Balcinrenone, a mineralocorticoid receptor modulator, is predicted to have similar clinical benefits as MRAs but with fewer safety concerns. This study evaluates if combining balcinrenone with dapagliflozin (an SGLT2 inhibitor) provides superior outcomes over dapagliflozin alone for patients with HF and impaired kidney function.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Dapagliflozin) * **Blinding:** Double-blind, double-dummy * **Randomization:** 1:1:1 Parallel-group

3.2 Planned Enrollment & Duration * **Target** **N**: 4,800 patients * **Number of Sites**: Approximately 700 sites in approximately 40 countries globally * **Estimated Study Start**: 12-Apr-2024 * **Estimated Primary Completion**: 11-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be ≥ 18 years old at the time of signing the informed consent. 2. A documented diagnosis of symptomatic HF (NYHA functional Class II to IV) with an LVEF assessment within the last 12 months. 3. Having had a recent HF event (hospitalization or urgent HF visit requiring parenteral medications) within six months prior to randomization. 4. Impaired kidney function with an eGFR of ≥ 20 to < 60 mL/min/1.73 m².

4.2 Exclusion Criteria 1. Systolic BP < 100 mmHg, or symptomatic hypotension within the past 24 hours at randomization. 2. Systolic BP ≥ 160 mmHg if on treatment with < 3 blood pressure-lowering medications or ≥ 180 mmHg irrespective of treatments. 3. Acute coronary syndrome (unstable angina or myocardial infarction), stroke, or transient ischemic attack within the previous 3 months. 4. Currently taking an MRA, or having specific contraindications to MRAs (such as hyperkalemia risk).

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency**: Once daily administration of one capsule and one tablet. Participants are randomized into one of three arms: * Balcinrenone/dapagliflozin 15 mg/10 mg capsule + matching placebo tablet * Balcinrenone/dapagliflozin 40 mg/10 mg capsule + matching placebo tablet * Dapagliflozin 10 mg tablet + matching placebo capsule * **Route of Administration**: Oral * **Duration of Treatment**: Event-driven. The estimated average study duration is 22 months (includes a screening period, an average 20-month blinded treatment period, and a 1-month follow-up on open-label dapagliflozin).

5.2 Concomitant & Prohibited Therapy * **Permitted**: Standard of Care (SoC) therapy for HF and impaired kidney function according to local guidelines. * **Prohibited**: MRAs; strong or moderate CYP3A4 inhibitors or inducers. Patients on prior open-label SGLT2 inhibitors must switch to blinded study therapy upon randomization and discontinue previous SGLT2i prescriptions.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to 1 month Informed consent,

medical history review, ECG, blood and urine tests, vital signs |
| **Baseline** | Day 0 | Randomization, switch from open-label SGLT2i
to blinded IP, first dose | | **Treatment Period** | Event-driven
(~20 months) | Continuous tracking for HF events, CV
hospitalizations, physical exams, QoL questionnaires | | **End of
Study** | Follow-up (+1 month) | Switch to open-label
dapagliflozin, final safety assessments, IP accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Reduction in the risk of CV death and HF events (hospitalizations or urgent HF visits). * **Measurement Method:** Event-driven, clinical tracking and independent central adjudication of cardiovascular and heart failure events.

7.2 Secondary & Exploratory Endpoints * **[Endpoint 1]** Evaluation of the safety profile of the combination therapy with special focus on renal markers and potassium levels. * **[Endpoint 2]** Improvement in patient clinical status, functional capacity, and overall quality of life.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Conducted via routine physical examinations, vital sign reviews, ECGs, and blood/urine laboratory tests checking on overall health.
 - **Serious Adverse Events (SAEs):** Tracking specific HF morbidities, worsening kidney function, severe hypotension, and hyperkalemia (potassium monitoring is critical due to standard MRA safety concerns).
 - **Data Monitoring Committee (DMC):** Independent safety monitoring over the anticipated ~38-month total global study timeline.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy endpoints (CV death and HF events). Safety Set for adverse events.
 - **Sample Size Justification:** The trial is event-driven. Approximately 4,800 patients are required to adequately power the study to detect a statistically significant superior clinical benefit of balcinrenone/dapagliflozin over dapagliflozin alone.
 - **Handling of Missing Data:** Standard time-to-event analysis methods; patients without an event will be censored at the date of their last known contact.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Global remote and onsite monitoring encompassing roughly 700 sites across 40 countries.
 - **Audit Trail:** eCRF completion for clinical events, concomitant medications, laboratory results, and central event adjudication.
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11. Study Identifiers & Governance

- **Primary ID:** D6402C00012
 - **Registry Number:** NCT06307652
 - **Sponsor/Lead Organization:** AstraZeneca
 - **Study Title:** Balanced-HF
 - **Official Regulatory Title:** A Phase III, Randomised, Double-blind Study to Evaluate the Effect of Balcinrenone/ Dapagliflozin, Compared With Dapagliflozin, on the Risk of Heart Failure Events and Cardiovascular Death in Patients With Heart Failure and Impaired Kidney Function
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12. Clinical Framework (Study Specific)

- **Primary Condition:** Chronic Heart Failure and Impaired Kidney Function
 - **Diagnosis Threshold:** NYHA Class II-IV; Impaired kidney function (eGFR ≥ 20 to < 60 mL/min/1.73 m^2)
 - **Medical Methodology:** Mineralocorticoid receptor modulator combined with an SGLT2 inhibitor
 - **Optimization Target:** Reduction in CV death and HF events without the severe hyperkalemia risks traditionally associated with standard MRAs
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13. Study Procedures & Methodology

- **Management Pathway:** Standard of Care (SoC) for HF + blinded investigational combination therapy
- **Education Protocol (HCP):** Site initiation and training regarding complex inclusion/exclusion criteria, including managing concomitant therapies and specific MRA contraindications
- **Education Protocol (Patient):** Informed consent, administration of study pills once daily, and completion of QoL surveys
- **Quality Audit Mechanism:** Tracking protocol adherence, especially successful transitions from prior SGLT2i prescriptions to blinded study drugs

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1. *Journal of the American Medical Association*, 1997; 277: 1039-1043.